

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

COMPUTER GRAPHICS

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SCOPE

2D ROTATION

2D Rotation overview

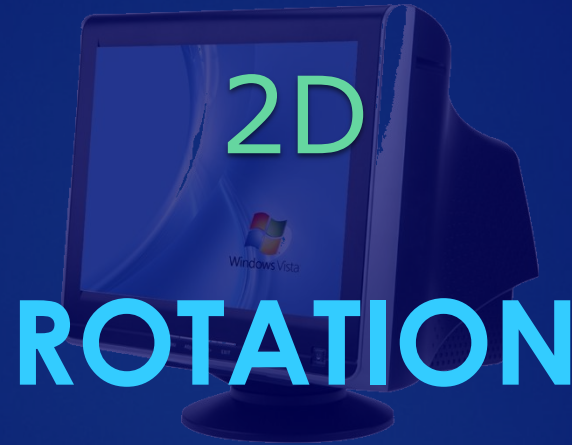
2D Rotation types

2D point of origin rotation overview

2D point of origin rotation formulas

Examples of 2D point of origin rotation

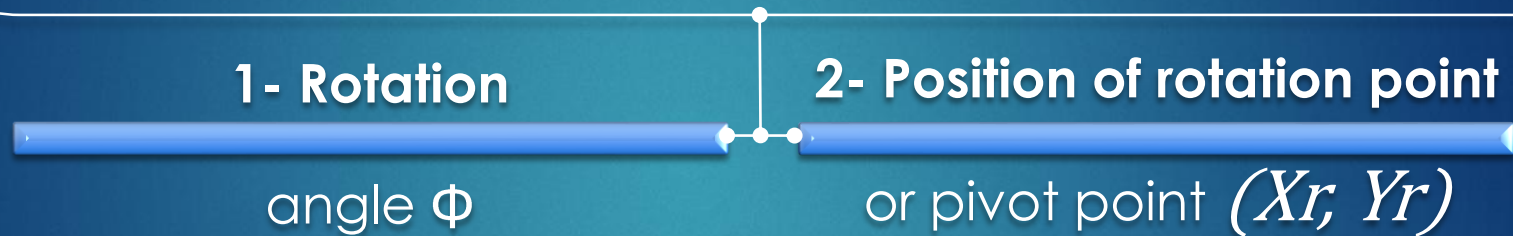




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2D ROTATION OVERVIEW?

A 2D rotation is applied to an object by repositioning it along a **circular path** in XY plane, to **rotate** the object **two parameters** are important, such as:



One or more objects or the entire image about any point in world space is possible to rotate in either **negative oriented** a (clockwise) or **positive oriented** (Anti-clockwise). There are two types of Rotate :

Rotation by Origin

Pivot Rotation



2D ROTATION Formula?

❑ Rotate about the origin formulas

$$\begin{aligned} X' &= x \cos(\theta) - y \sin(\theta) \\ Y' &= x \sin(\theta) + y \cos(\theta) \end{aligned}$$

Is used transformation that rotate a point an angle (θ) about the origin in the **Anti-clockwise** direction.

$$\begin{aligned} X' &= x \cos(\theta) + y \sin(\theta) \\ Y' &= -x \sin(\theta) + y \cos(\theta) \end{aligned}$$

Is used transformation that rotate a point an angle (θ) about the origin in the **clockwise** direction.

➤ Formula as a matrix

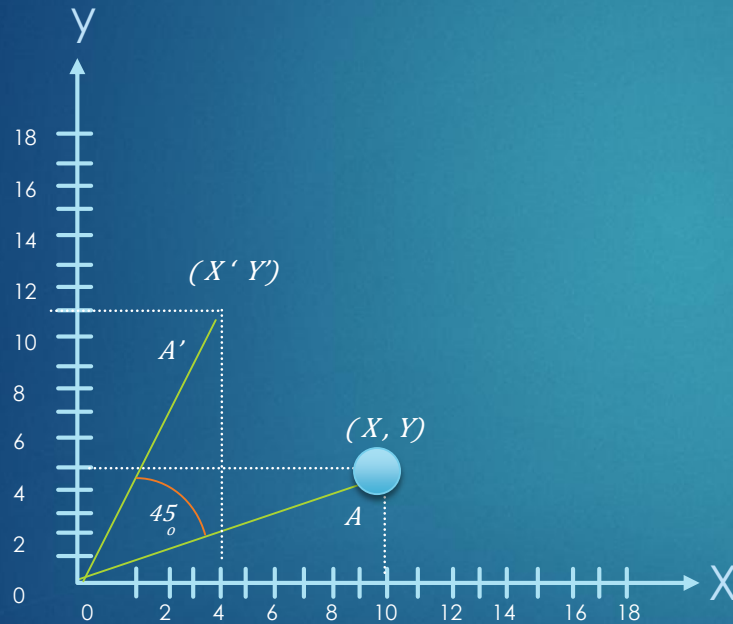
$$P' = P * R$$

$$\begin{bmatrix} X' & Y' \end{bmatrix} = \begin{bmatrix} X & Y \end{bmatrix} * \begin{bmatrix} \cos\Phi & \sin\Phi \\ -\sin\Phi & \cos\Phi \end{bmatrix}$$



EXAMPLE OF ROTATION

- ✓ Rotate a point with a coordinate of $A (10,5)$ into 45 degree angle anticlockwise.



$$[X' \ Y'] = [X \ Y] * \begin{bmatrix} \cos\Phi & \sin\Phi \\ -\sin\Phi & \cos\Phi \end{bmatrix}$$

$$[10 \ 5] * \begin{bmatrix} \cos(45) & \sin(45) \\ -\sin(45) & \cos(45) \end{bmatrix}$$

$$[10 \ 5] * \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$[x' \ y'] = \left[\frac{10}{\sqrt{2}} - \frac{5}{\sqrt{2}} \quad \frac{10}{\sqrt{2}} + \frac{5}{\sqrt{2}} \right]$$

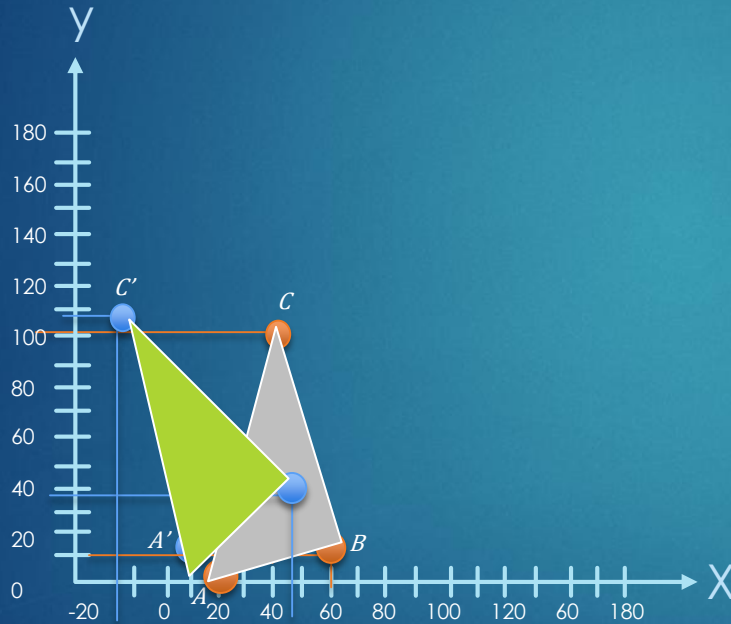
$$\begin{bmatrix} \frac{5}{\sqrt{2}} & \frac{15}{\sqrt{2}} \end{bmatrix}$$

$$P' = (4, 11)$$



EXAMPLE OF TRANSLATION

- ✓ A triangle with vertices **(20,0)**, **(60,10)**, and **(40,100)** been rotated **30° counter clockwise**. What are the new coordinates of this triangle after Rotation. .



$$X' = x \cos(\theta) - y \sin(\theta) \quad Y' = x \sin(\theta) + y \cos(\theta)$$

$$A' = \begin{array}{l} 20 * 0.866 - 0 = 17 \\ 20 * 0.5 + 0 = 10 \end{array}$$

$$A' = (17, 10)$$

$$B' = \begin{array}{l} 60 * 0.866 - 5 = 47 \\ 60 * 0.5 + 8.66 = 39 \end{array}$$

$$B' = (47, 39)$$

$$C' = \begin{array}{l} 40 * 0.866 - 50 = -15 \\ 40 * 0.5 + 86.66 = 107 \end{array}$$

$$C' = (-15, 107)$$





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YOU PERFECT**





Always Seek Knowledge



THE END

